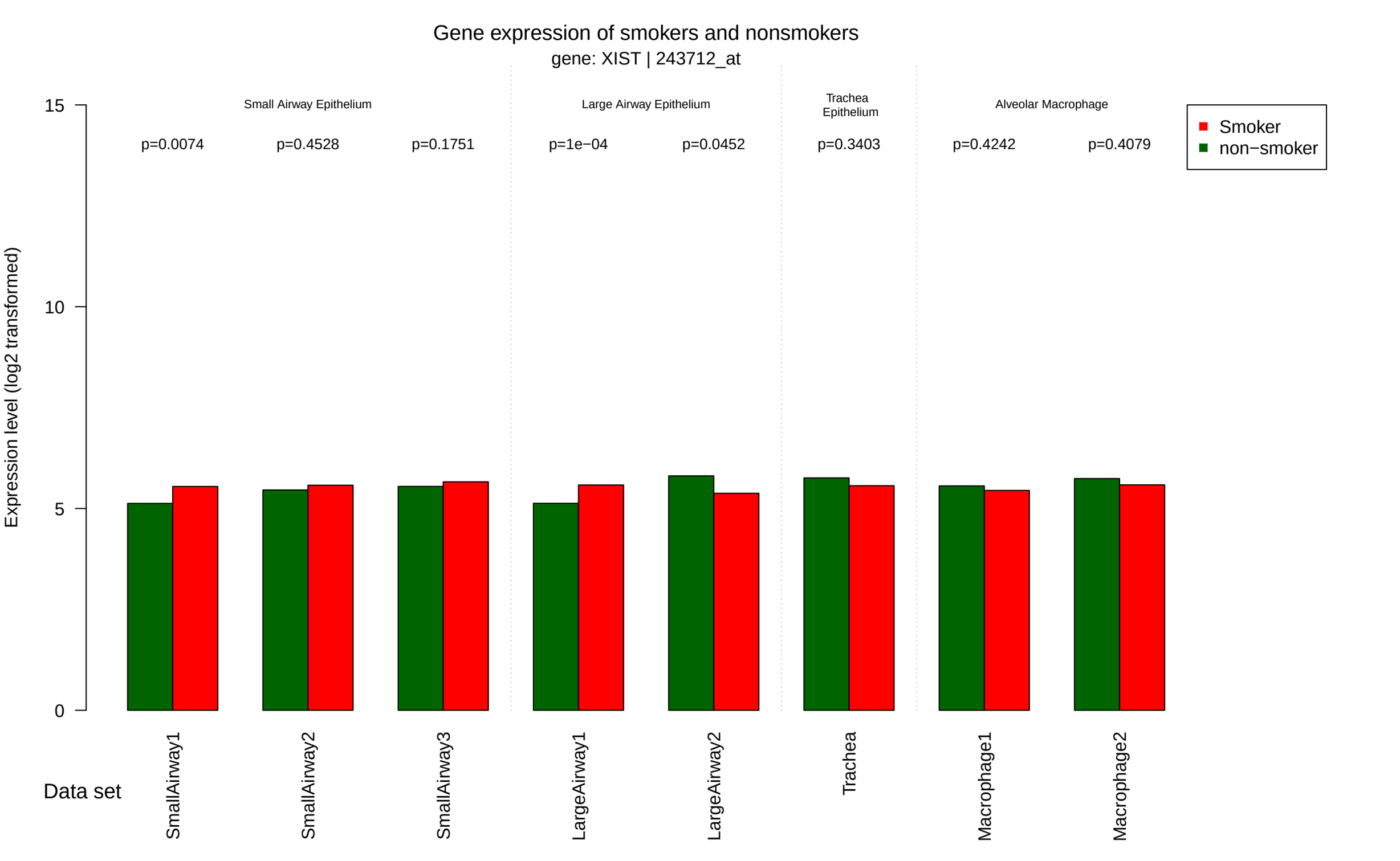
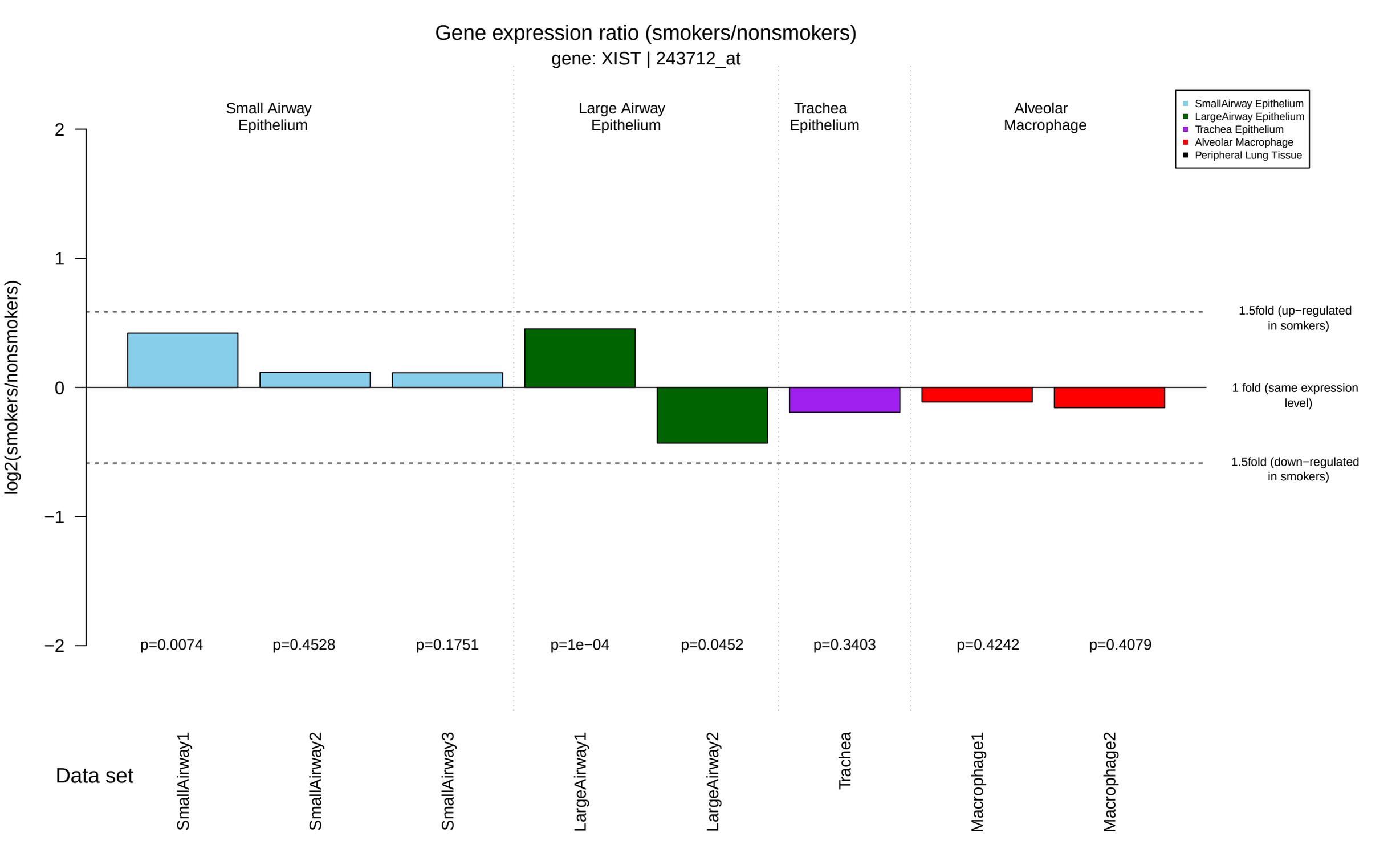
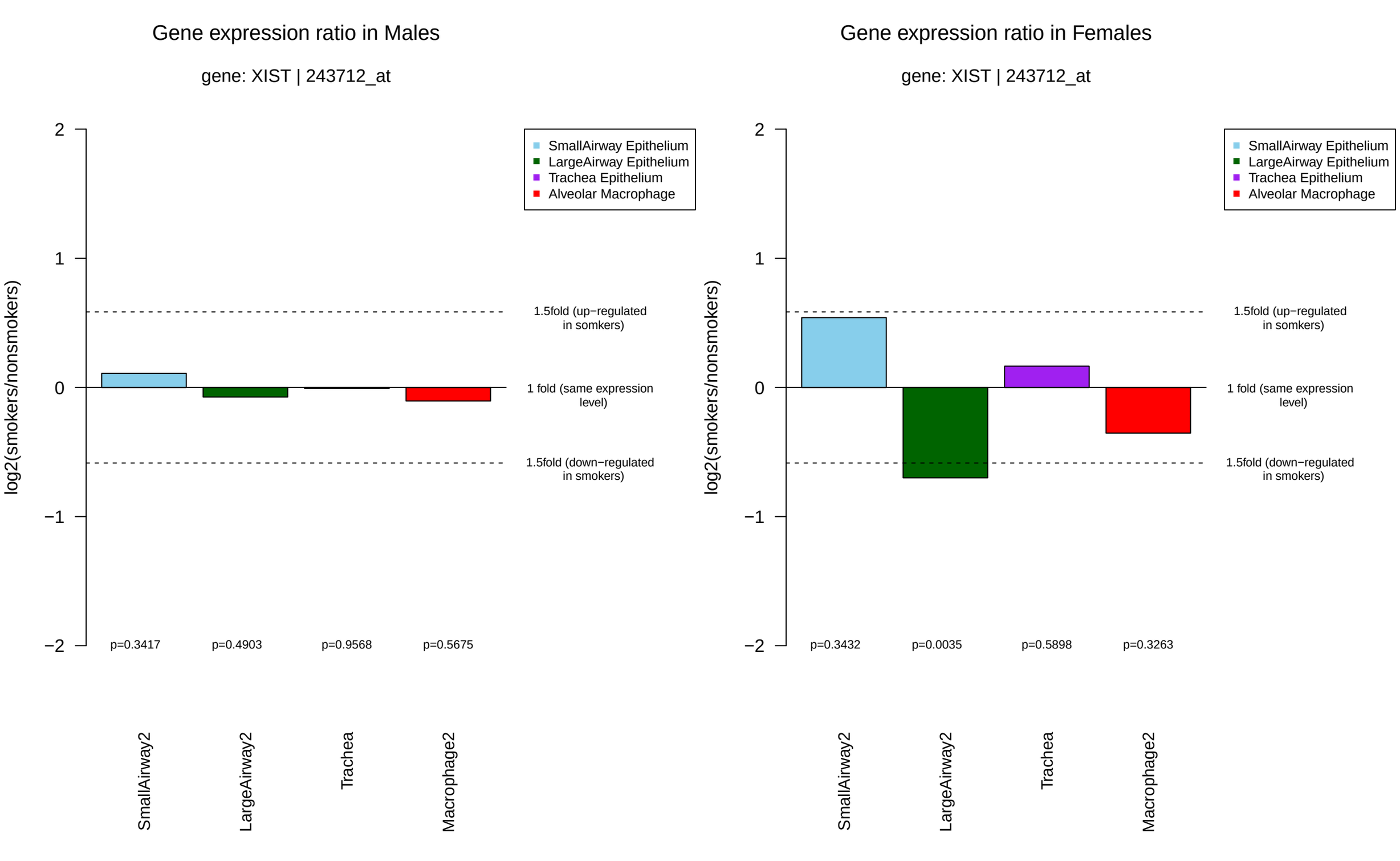
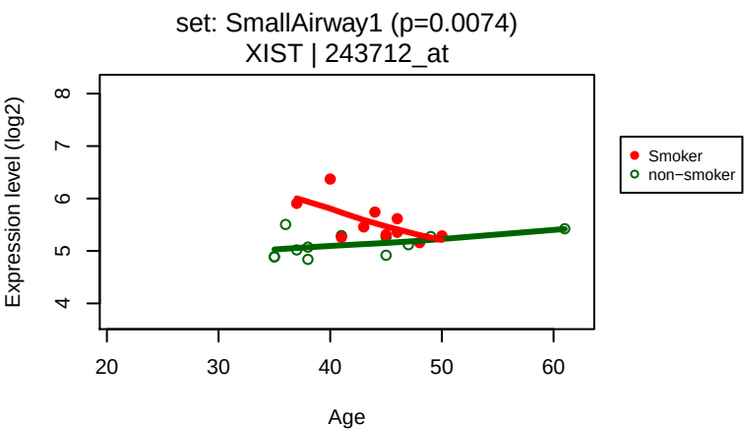
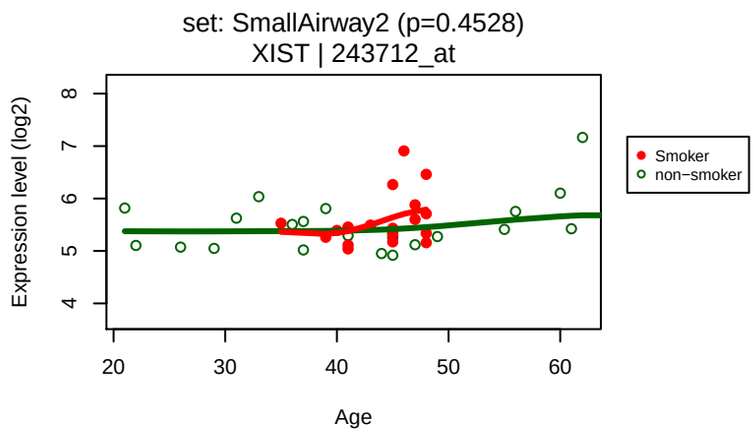
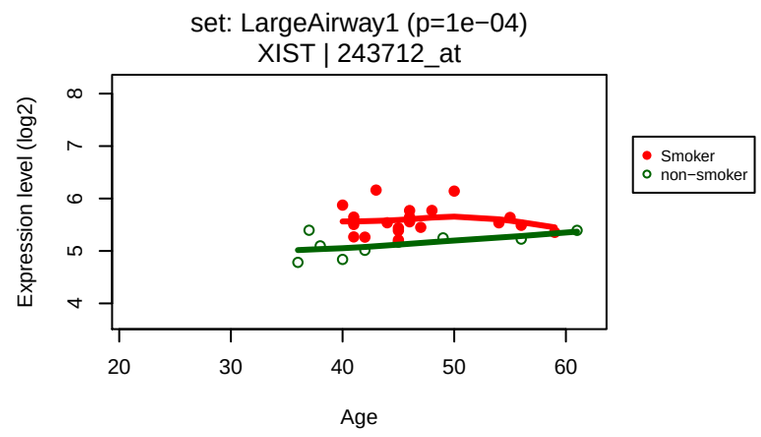

Gene expression of smokers and nonsmokers
gene: XIST | 243712_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0074
p=0.4528
p=0.1751
p=1e−04
p=0.0452
p=0.3403
p=0.4242
p=0.4079
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

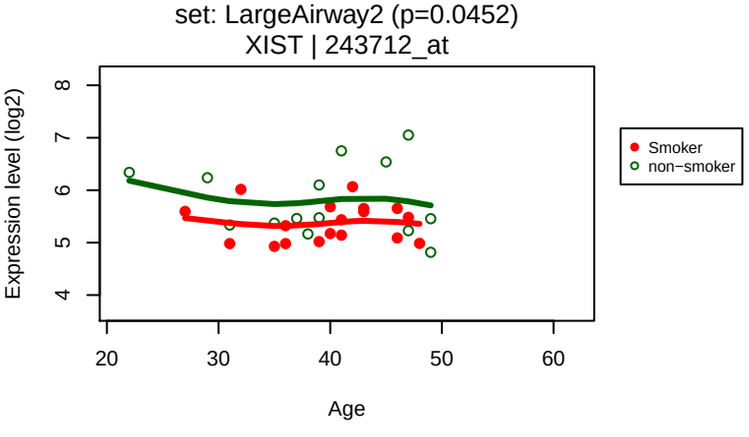

Gene expression ratio (smokers/nonsmokers)
gene: XIST | 243712_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0074
p=0.4528
p=0.1751
p=1e−04
p=0.0452
p=0.3403
p=0.4242
p=0.4079
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

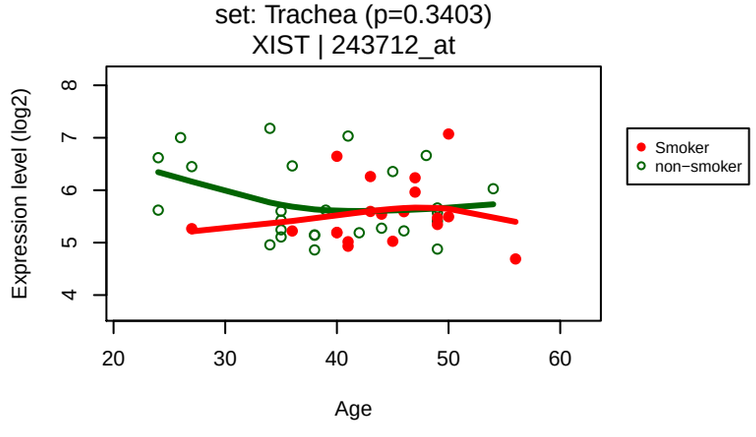

Gene expression ratio in Males
gene: XIST | 243712_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3417
p=0.4903
p=0.9568
p=0.5675
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: XIST | 243712_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3432
p=0.0035
p=0.5898
p=0.3263
SmallAirway2
LargeAirway2
Trachea
Macrophage2

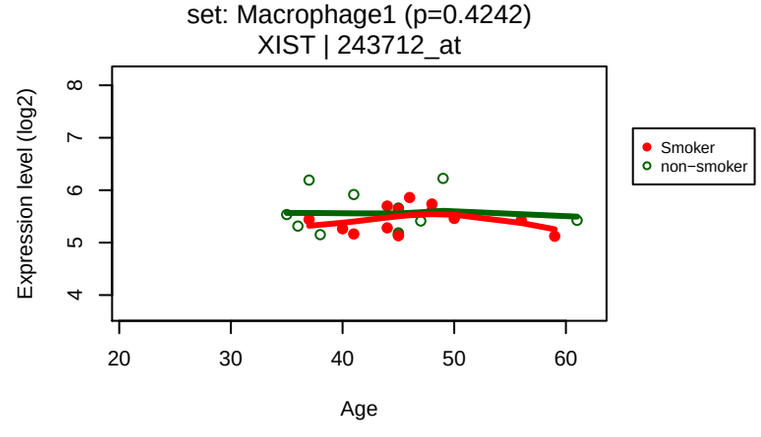

set: SmallAirway1 (p=0.0074)
XIST | 243712_at
Expression level (log2)
Age
Smoker
non−smoker

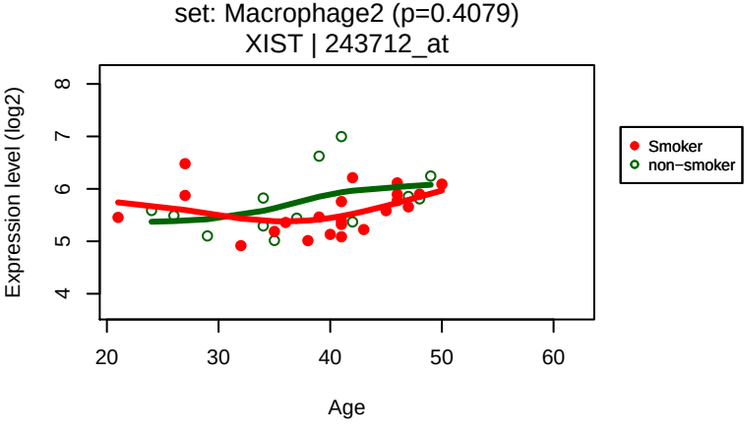

set: SmallAirway2 (p=0.4528)
XIST | 243712_at
Expression level (log2)
Age
Smoker
non−smoker

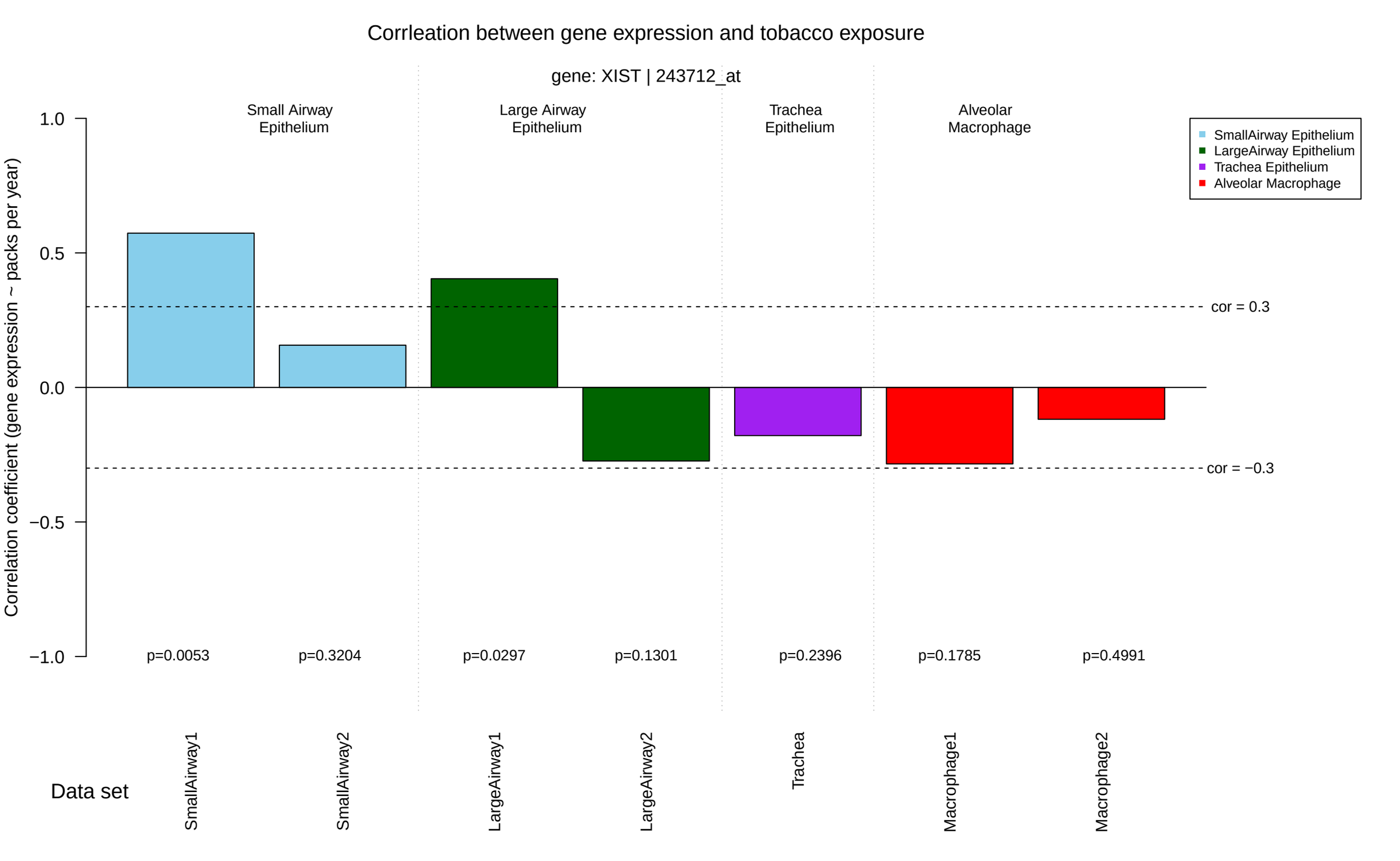

set: LargeAirway1 (p=1e−04)
XIST | 243712_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0452)
XIST | 243712_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3403)
XIST | 243712_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4242)
XIST | 243712_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4079)
XIST | 243712_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: XIST | 243712_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0053
p=0.3204
p=0.0297
p=0.1301
p=0.2396
p=0.1785
p=0.4991
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2